Personal, Background, and Future Goals Statement

Do you recall the big "E" on the first line of Snellen chart when they administered an eye exam for determining visual acuity? The big "E" is special to me because it is the only letter that I am capable of accurately detecting when I am standing two hundred feet away. My current vision is 20/200 and cannot be corrected with standard prescription glasses. I have a juvenile form of macular degeneration called X-Linked Retinoschisis. I fall under a narrow category of the visual spectrum in which I am not completely blind nor perfectly sighted. Despite this detrimental sensory impairment affecting my retinal neurons, I still have enough functional vision to comprehend and perform exceptional scientific research.

Background

My vision counselor told me to be realistic in pursuing a career path because most visually impaired and blind students fail to make it in the life sciences. My confidence was completely stripped away, and I dropped out of university nine years ago since I could no longer afford to meet my basic needs along with the lack of support from an institution that was designed to aid the visually impaired. At the time, it was the best decision of my life considering I didn't have a purpose for studying. I only enrolled in university because everyone else did the same. I was just following the normative basic timeline. I ended up working different jobs and exploring my career path. I experienced working at a warehouse, a kitchen, and completed an internship in information technology. Even though I did not enjoy these working experiences, I gained valuable practical transferrable skills that would eventually help me in the sciences. I learned how to communicate, collaborate with teams, follow a protocol, take initiative, and rebuild my confidence.

Eventually, after two years of working full time, I decided to enroll in my local community college when I heard that the tuition was free for residents. This was a sign for me to pursue opportunities in higher education that I couldn't before, due to the limited resources available to me. I enrolled in the biotechnology program at my college because it was one of the only programs that was designed for working adults. I took classes at night and worked during the day. The moment I realize when I wanted to truly do science was when I first saw green fluorescent protein in hamster kidney cells that would not normally be expressed. The method of taking a mechanism that is not normally present within an organism and engineering them to perform unnatural function really showed me the power of science and the potential impacts it could have. The biotech program taught me important research skills beyond performing lab techniques such as reading papers, analyzing statistics, and critical thinking of why each step is performed.

Intellectual Merit

After completion of my coursework, my professor encouraged me to apply for the California Institute of Regenerative Medicine (CIRM) internship program at the City College of San Francisco. ***I was one of the few applicants who was accepted into the CIRM stem cell internship program***. I decided to pursue research in translational science in cell therapy because I wanted to make a direct impact. My first full time research experience was at the Vitalant Research Institute under Dr. Muench in the stem cell therapy core. I carried on multiple projects working in the lab. One of my projects was to optimize methods for enumerating stem cell frequencies. The colony forming unit (CFU) is a common assay used for enumerating stem cell frequency, but this method was ineffective for counting primary mesenchymal stem cell lines because of their in vitro migrative properties. Due to their stem cell migrative properties, I kept getting inaccurate replicate cell counts. I decided to compare the CFU against another enumeration method. I used the extreme limiting dilution assay (ELDA) as an alternative system to measure the accuracy of enumerating stem cell frequency. ***I determined that the ELDA was more effective and accurate in determining mesenchymal stem cell populations versus the CFU assay***. After optimizing the ELDA, I used the assay to determine stem cell frequencies of frozen samples in additional collaborative work.

In addition, my lab also worked in collaboration with a biotech start-up company called X-Therma. Their ultimate goal was to develop a cryoprotectant to prolong the viability and shelf life of organs during the process of organ procurement for transplantation. However, before X-Therma can begin their aim for freezing whole organs, they need to develop formulations that are capable of freezing single cell

suspensions. X-Therma was synthesizing novel cryoprotectants that mimicked the properties of antifreeze proteins in fish that allowed them to survive in the Antarctic at below freezing temperature. This cryoprotectant product was being developed because the current methods for freezing stem cells during transplantation requires 10% dimethyl sulfoxide, which are toxic to cells and reduce their viability after post thawing. ***I assisted my mentor in narrowing down the most effective cryoprotectant formulations on primary cell lines used in a study that was published in our lab by Gilfanova Et. al, 2021***. I tested the proliferation of stem cell regeneration after post thawing them from different formulations. This experience gave me validation as a scientist because I was able to achieve comparable viable results similar to my mentor's data using different methods. This training was valuable for me as this was the moment that I knew wanted to continue to pursue academic research. I learned that being a scientist requires more than just scientific knowledge. For example, when methods failed to work, I had to go back to the literature and understand why they did not work and find alternative methods. To be a well-rounded scientist requires collaboration, writing, communication, troubleshooting, creative thinking, data analysis, and being open to new methods. ***After completion of my internship, I received my associate degree in biotechnology***.

I decided to continue my bachelor's degrees in cell and molecular biology at San Francisco State University. ***I applied and received an academic scholarship from the Genentech Foundation that gave me a stipend and covered my tuition expenses for my undergraduate degree in exchange for performing research on campus.*** This allowed me to quit my job and focus part time doing research while taking classes. I wanted to continue to pursue stem cell research, but my institution had limited resources and did not offer stem cell research, so I decided to go into vision research because I had always wanted to know the molecular mechanism for how vision operates.

In my second lab, I work under Dr. Anastassov in retinal connectomics. The lab focused on comparative biology of simplex (1 type of photoreceptor) versus duplex (2 types of photoreceptors) visual systems. The simplex visual system is being studied in our lab because it has the potential to lead to new research approaches and knowledge of visual systems. My project in the lab was to take serial sections of electron microscopy data and 3D reconstruct organelles. I analyzed the morphology and structures of mitochondria and ribbon synapses in the simplex retina. ***I discovered that photoreceptors ribbon synapses from the simplex visual system had consistent volume, surface areas, and docked synaptic vesicles despite their differences in the number of quantities and sizes.*** The simplex visual system had rarely been explored and not well understood and this discovery may perhaps explain that the simplex visual system requires a constant rate for vesicle fusion in lateral feedback inhibition. In addition, I investigated the complexity of mitochondrial morphology and how they differ from neuronal cell types of the simplex retina. In my assigned next project, I will be gathering reconstructions of invaginating postsynaptic cell processes downstream photoreceptors to understand the ratio of connecting interneurons cells. ***I am in the process of finalizing data and will eventually write an optional senior thesis on my research findings for my department.***

My research trajectory for stem cell biology to vision science completely changed when ***I got accepted a conference at the Association for Research in Vision and Ophthalmology to present my research findings.*** I presented my findings from Dr. Anastassov's lab on the structural diversity of 3D reconstructions of mitochondria and quantifications of synaptic vesicles in the simplex retina. At the conference, I met a lot of inspiring and passionate scientists working on visual systems and I was exposed to research that had sparked my interest. I never saw the eye as an inspirational tool because of my limited vision but vision science changed my perspective and really opened my eyes. The eye is a gold mine, and I could see endless opportunities. This conference helped me envision the type of science that I eventually would like to pursue in the future. My work in the retina was recognized and ***the faculty in my department awarded me the National Institute for Health Undergraduate Research Training for Student Enhancement program***.

Broader Impacts

A lot of modern-day technology that you see today were inspired by blind and low vision communities. For example, the first typewriter was invented for the blind and was eventually transformed into the keyboard. Unfortunately, the blind and visually impaired have been historically underrepresented when it comes to the life sciences. In fact, the percentage of blind and visually impaired students was practically nonexistent in the biological sciences when I was enrolled throughout college. Science is already challenging enough but imagine adding another layer on top of learning science. Doing science without sight or limited sight increases the difficultness by tenfold. Challenges in sciences does not even account for obstacles in life outside of education. In essences, the blind and visually impaired population must put in more effort and go the extra mile to succeed in higher education. Consequently, the underrepresentation, limited access to accessible technology, and lack of mentoring puts us at an even greater disadvantage.

From experience, many professors in the sciences are inept when it comes to teaching blind or low vision students. How do I succeed in an education system that was not designed for the blind? I had to develop my own techniques and work with professors to navigate around the lab. Being a student is like performing cell culture. Under the right conditions and growth factors, the cells can thrive and proliferate in their environment. However, if you place the cells under the wrong medium, the cells may become senescent or even lose their viability. I was learning as I went, adjusting my own growth medium along the way. Finding out what method worked best for me by utilizing adjustable magnifiers, portable closed-circuit television, wearable technologies, flashlight on my smartphone, computer accessibility features. I had to learn how to advocate for myself to acquire assistive technology. These were all things I wished I could have gotten advice from someone else to make my life easier. My journey in science was lonely because I did not have anyone else that I could share these experiences with. This is why ***I have done outreach at Vista Center for the Blind and Visually Impaired and spoke to vision impaired high school students encouraging them to pursuing science.*** I want to encourage other low vision students that it is not impossible to pursue a career in the life sciences.

The reason why I have gotten this far in science was because I had amazing mentors. I believe that having a mentor in any field is important and seeing someone else with similar circumstances gives encouragement. I mentored students in both my academic research experience and teaching experience. I took time out of my projects to train new colleagues in the lab how to extract bone marrow, maintain mouse colonies, flow cytometry, start primary cell lines, and perform lab calculations. ***I taught as a supplemental instructor for a general biology course as an undergraduate student***. I prepared curriculum that helped first year students engage and retain science information. Science is difficult but I ensure that the knowledge I acquired from my previous mentors gets passed down to the next generation of scientists through analogies and real-life connections that are memorable.

There are very few blind and visually impaired people in the life sciences who can give practical advice to navigate the field. I don't want to just do science; I want to transform science and make it more accessible for the visually impaired and blind. I want to be able to pioneer new methods and technologies that allow the visually impaired community to participate in science. Science without leaders that are blind and visually impaired discourages visually impaired students to pursue a career in the life sciences. I believe that excluding the visually impaired and blind population from science is a huge loss for the scientific community because when I look at the blind and visually impaired community, I see a source of untapped innate curiosity, innovation, and creativity waiting to be utilized.

Future Goals

I want to pursue a PhD because I want to go into industry research and development. My primary goal is to help advance and develop high quality scalable products that can help with improving sight restoration. Improving sight restoration by a few percent can significantly increase the quality of life for those with no light perception or vision impairments. I am looking to refine my skills as an independent researcher through a PhD in retinal regeneration. I want to advance my critical thinking abilities in the scientific process and take charge of my own projects. With the assistance from the NSF GFRP, I believe that I have to potential to advance scientific knowledge and make science more accessible.